Preface

Simone Azeglio SIMONE.AZEGLIO@GMAIL.COM
Institut de la Vision & École Normale Supérieure, Paris
Bahareh Tolooshams BTOLOOSH@CALTECH.EDU
California Institute of Technology
Chase van de Geijn CHASEVDGEIJN@GMAIL.COM
University of Amsterdam
Christian Shewmake SHEWMAKE@BERKELEY.EDU
University of California, Berkeley
Sophia Sanborn SOPHIA.SANBORN@GMAIL.COM
Stanford University & Enigma Project
Nina Miolane NINAMIOLANE@UCSB.EDU
University of California, Santa Barbara

Editors: Simone Azeglio, Bahareh Tolooshams, Chase van de Geijn, Christian Shewmake, Sophia Sanborn, Nina Miolane

1. Symmetry and Geometry in Neural Representations

A growing body of work in computational neuroscience points toward a unifying account of the neural code. Across sensory and motor regions of the brain, neural circuits are found to mirror the geometric and topological structure of the systems they represent—either in their synaptic connectivity, or in the implicit manifold traced out by their activity. The phenomenon appears in the ring of neurons encoding head direction in the fly, in the toroidal structure of grid cell population activity, and in the low-dimensional manifolds observed in motor cortex. Together these findings suggest a general computational strategy, employed throughout the brain, of preserving the geometric structure of data across successive stages of information processing.

Independently but convergently, the same strategy has taken hold in deep learning. Geometric deep learning builds geometric priors directly into artificial neural networks so that the structure of a signal is preserved as it passes through the layers of a model, with demonstrable gains in computational efficiency, robustness, and generalization.

That these two lines of inquiry have converged suggests deep, substrate-agnostic principles for information processing. Symmetry and geometry were instrumental in unifying our models of the fundamental forces and elementary particles in twentieth-century physics. They may prove equally instrumental in revealing how neural systems—biological and artificial alike—form useful representations of the world.

The 2024 edition took world models as its particular focus. An agent that models its environment has to represent more than the structure of a single observation; it must also represent how that structure changes under the agent's own actions and over time. This is where symmetry and geometry become directly relevant: a representation organized around the transformations a world admits can predict by moving along known orbits instead of memorizing trajectories, which makes it cheaper to learn and easier to transfer. The question of how a representation acquires structure is then inseparable from the dynamics that structure must support. That relationship was the subject of the workshop's panel on world models and equivariance.

2. The Workshop

The third annual NeurIPS Workshop on Symmetry and Geometry in Neural Representations (NeurReps) was convened to bring together researchers working at the intersection of applied geometry, deep learning, and neuroscience, with the aim of advancing this understanding and drawing out geometric principles for neural information processing. We hope that this venue, and the community that has formed around it, will continue to support the development of a geometric approach to neural representations while strengthening ties to the mathematics community. The Neural Information Processing Systems conference itself emerged historically from theoretical neuroscience and connectionism, and the workshop is intended in part to reinforce that bond.

2.1. Call for Papers

Our call for papers invited submissions for publication in this volume and for presentation at the workshop. It requested novel research at the intersection of geometric deep learning, computational neuroscience, geometric statistics, and topological data analysis—work incorporating symmetry, geometry, or topology into the design of artificial neural networks, the analysis of neural data, or theories of neural computation.

The call yielded 84 submissions, which were reviewed double-blind. Our 158 reviewers produced 231 reviews, which were aggregated and assessed by the area chairs and editors for final inclusion. This resulted in 61 accepted works: 16 full-length papers and 45 extended abstracts. All 16 full-length papers are published in this volume, by choice of the authors. Reviews of the accepted works can be found on the NeurReps OpenReview portal.

2.2. The Schedule

The workshop was held on Saturday, December 14th, 2024 in West Ballroom C of the Vancouver Convention Center, Vancouver, British Columbia, Canada. It featured five invited talks spanning neuroscience, machine learning, geometry, and their intersection, with a particular focus this year on world models, together with a discussion panel. All accepted works were presented as posters during the workshop, with a selection of submissions chosen for oral presentation. Simone Azeglio delivered the opening remarks on behalf of the organizing committee, and the day closed with announcements from the committee.

Invited Talks

- *Geometry of the Distribution of Natural Images*
  Eero Simoncelli, New York University & Flatiron Institute
- *The Cortical Topography of Visual Space Representation*
  Talia Konkle, Harvard University
- *In-Context Symmetries*
  Stefanie Jegelka, MIT & TU Munich
- *Reverse-engineering cognitive and neural representations with multilevel computational theories*
  Ilker Yildirim, Yale University

- *Simulating the brain and body of the fruit fly*
  Srini Turaga, HHMI Janelia Research Campus

Discussion Panel: World Models & Equivariance Panelists: Sophia Sanborn, Eero Simoncelli, Talia Konkle, Ilker Yildirim, Srini Turaga, Stefanie Jegelka, and Savannah Thais (Columbia University).

Contributed Talks

- *ManiPose: Manifold-Constrained Multi-Hypothesis 3D Human Pose Estimation* — Cédric Rommel
- *Communication subspaces align with training in ANNs* — Peter Poggi
- *Leveraging Symmetry to Accelerate Learning of Trajectory Tracking Controllers for Free-Flying Robotic Systems* — Pratik Kunapuli, Nishanth Rao
- *Exploring Geometric Representational Alignment through Ollivier-Ricci Curvature and Ricci Flow* — Nahid Torbati
- *Does Equivariance Matter at Scale?* — Johann Brehmer
- *BiEquiFormer: Bi-Equivariant Representations for Global Point Cloud Registration* — Stefanos Pertigkiozoglou, Evangelos Chatzipantazis
- *Harmformer: Harmonic Networks Meet Transformers for Continuous Roto-Translation Equivariance* — Tomáš Karella
- *Probabilistic Nested Homogeneous Spaces for Dimensionality Reduction* — Xiran Fan
- *Visualizing Loss Functions as Topological Landscape Profiles* — Michael W. Mahoney
- *Topological Blindspots: Understanding and Extending Topological Deep Learning Through the Lens of Expressivity* — Yam Eitan
- *Enhancing the Expressivity of Temporal Graph Networks through Source-Target Identification* — Benedict Aaron Tjandra
- *Efficient Subgraph GNNs via Graph Products and Coarsening* — Guy Bar-Shalom

3. The Community

To support the growth of this research area beyond the annual workshop, we maintain a digital community for NeurReps, which at the time of writing has over 1200 members. Instructions for joining and contributing can be found on the community page of our website.

4. Program Committee

We extend our gratitude to our area chairs, who supervised the reviewing process and led the selection of contributed talks, and to our 158 reviewers, whose collective efforts produced the 231 reviews on which this volume rests. Their dedication and expertise have been integral to the success of the workshop. Thank you.

Area Chairs

- Maurice Weiler — Algebra & Geometry
- Alex Williams — Neuroscience & Interpretability
- Mustafa Hajij — Topology & Graphs

Reviewers

- Abhinav Kumar
- Abiy Tasissa
- Adele Lantow
- Adrish Dey
- Aishwarya Balwani
- Alessandro Salatiello
- Alexandros Dimitrios Keros
- Ali Parviz
- Alison Pouplin
- Alpha Renner
- Alvin Jin
- Andreea Deac
- Ankita Shukla
- Antonio Briola
- Anupam Chaudhuri
- Arash Behboodi
- Arif Dönmez
- Arjun Karuvally
- Arnab Ray
- Artem Moskalev
- Arturs Berzins
- Aslan Satary Dizaji
- Bahareh Tolooshams
- Behrooz Tahmasebi
- Benjamin Aslan
- Bilal Alsallakh
- Binxu Wang
- Bo Zhao
- Bongjin Koo
- Charles Godfrey
- Chase van de Geijn
- Cheng Xin
- Chi Xie
- Claire Donnat
- Congyue Deng
- Daniel Herbst
- Daniel Herrera-Esposito
- Daniel Platt
- David A. R. Robin
- David Robert Reich
- Dehong Xu
- Demba E. Ba
- Demi Qin
- Derek Lim
- Dongmian Zou
- Edward De Brouwer
- Eloy Geenjaar
- Emanuele Rodolà
- Eric Nalisnick
- Eric Qu
- Erik J Amezquita
- Erik J Bekkers
- Fabian H. Sinz
- Francesco Calisto
- Frederic BARBARESCO
- Gavin Kerrigan
- Gergely Berczi
- Guillaume Huguet
- Hamza Keurti
- Jacob A Zavatone-Veth
- Javier Gonzalvo
- Jens Agerberg
- Jianke Yang
- Johan Mathe
- John J. Vastola
- Jonas Kgomo
- Joshua Southern
- Jung Yeon Park
- Juyong Lee
- Kamesh Krishnamurthy
- Karn Tiwari
- Katharina Limbeck
- Kijung Yoon
- Konstantin Friedrich Willeke
- Lama Moukheiber
- Lazar Supic
- Leander Girrbach
- Liu Ziyin
- Long Tran-Thanh
- Lorenzo Posani
- Maghesree Chakraborty
- Manos Theodosis
- Manu Madhav
- Maosheng Yang
- Marco Guerra
- Marco Pegoraro
- Marco Piangerelli
- Matt Thomson
- Matthew Farrell
- Matthias H. Hennig
- Maximilian Thiessen
- Michael G. Rawson
- Mustafa Hajij
- Nello Blaser
- Nicola Toschi
- Nima Dehmamy
- Oguzhan Keskin
- Paul Samuel Ignacio
- Petar Veličković
- Peter Bentley
- Qingsong Wang
- Ramakrishnan Iyer

- Rana Muhammad Shahroz Khan
- Rik Sarkar
- Rishabh Anand
- Rishi Sonthalia
- Rucha Bhalchandra Joshi
- Salvish Goomanee
- Samuele Papa
- Samy Blusseau
- Santiago Velasco-Forero
- Sara Scaramuccia
- Sékou-Oumar Kaba
- Semih Cantürk
- Seunghyuk Cho
- Sharvaree Vadgama
- Shreyas N. Samaga
- Shubham Choudhary
- Siba Smarak Panigrahi
- Sigurd Gaukstad
- Soham Mukherjee
- Sourabh Palande
- Sree Harsha Tanneru
- Stephan Chalup
- Sungjun Cho
- Takehiro Aoshima
- Tamal K. Dey
- Thomas Gebhart
- Timothy Doyeon Kim
- Tobias Cheung
- Tom Needham
- Tomas Karella
- Uri Cohen
- Valentin Duruisseaux
- Vasco Portilheiro
- Victoria Klein
- Vincent Tao Hu
- Wei Ye
- Wenhao Zhang
- Wolfgang Polonik
- Yan Yan
- Yanan Long
- Ying Nian Wu
- Yivan Zhang
- Yuzhou Chen
- Zach Cohen
- Zachary Lubberts